Curso:

1. Utilizar la técnica de mínimos cuadrados para ajustar los siguientes datos

x	-2	-1	0	1	2
y	-3	-1	0	3	5

mediante una función de la forma $y = ax + b$.

2. Determinar el conjunto de todos los $x \in \mathbb{R}^3$ tales que $A^{1961}x = [2 \quad 0 \quad 2]^T$, donde

$$a(-2) + b = -3$$
$$a(-1) + b = -1$$
$$a(0) + b = 0$$
$$a(1) + b = 3$$
$$a(2) + b = 5$$

$$\begin{aligned} -2a + b &= -3 \\ -a + b &= -1 \\ b &= 0 \\ a + b &= 3 \\ 2a + b &= 5 \end{aligned}$$

$$\begin{pmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

Aplico $A^T A \hat{x} = A^T b$

$$\begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} -2 & -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -3 \\ -1 \\ 0 \\ 3 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} 10 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} \hat{a} \\ \hat{b} \end{pmatrix} = \begin{pmatrix} 20 \\ 4 \end{pmatrix}$$

$$\begin{cases} 10\hat{a} = 20 & \rightarrow \boxed{\hat{a} = 2} \\ 5\hat{b} = 4 & \rightarrow \boxed{\hat{b} = \frac{4}{5}} \end{cases}$$

$$\boxed{y = 2x + \frac{4}{5}}$$

2. Determinar el conjunto de todos los $x \in \mathbb{R}^3$ tales que $A^{1961}x = \begin{bmatrix} 2 & 0 & 2 \end{bmatrix}^T$, donde

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

Dado

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

I) Calculo los autovalores

$$p_A(\lambda) = \det \begin{pmatrix} 1-\lambda & 1 & 0 \\ 1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{pmatrix}$$

$$= (1-\lambda)^2 (2-\lambda) - [(1-\lambda)(1-\lambda)]$$

$$= (1-\lambda)(2-\lambda)(1-\lambda) - [0 + (1-\lambda) + (1-\lambda)]$$

$$= (1-\lambda)^2 (2-\lambda) - 2(1-\lambda)$$

$$= (1-\lambda)[(1-\lambda)(2-\lambda) - 2]$$

$$= (1-\lambda)[2 - \lambda - 2\lambda + \lambda^2 - 2]$$

$$= (1-\lambda)[\lambda^2 - 2\lambda - \lambda]$$

$$= (1-\lambda)[\lambda^2 - 3\lambda]$$

$$= (1-\lambda)\lambda(\lambda-3)$$

$\boxed{\lambda_1 = 1}$ $\boxed{\lambda_2 = 0}$ $\boxed{\lambda_3 = 3}$

Busco los autoespacios de cada autovalor

$$S_1 = \text{Nul}\begin{pmatrix} 1-1 & 1 & 0 \\ 1 & 2-1 & -1 \\ 0 & -1 & 1-1 \end{pmatrix} \rightsquigarrow \text{Nul}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\text{Nul}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$x_2 = 0$$

$$x_1 + x_2 - x_3 = 0$$

$$x_1 - x_3 = 0 \Rightarrow \boxed{x_1 = x_3}$$

$$[x_1 \quad x_2 \quad x_3]^T = [x_1 \quad 0 \quad x_1]$$

$$= x_1 [1 \quad 0 \quad 1]^T$$

$$\boxed{S_1 = \text{gen}\{[1 \quad 0 \quad 1]^T\}}$$

Busco $S_0 = \text{Nul} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \sim \text{Nul} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 - x_3 = 0 \end{cases} \rightarrow \boxed{x_2 = -x_1}$$

$$-x_1 - x_3 = 0 \rightarrow \boxed{x_3 = -x_1}$$

$$[x_1 \quad x_2 \quad x_3]^T = [x_1 \quad -x_1 \quad -x_1]^T$$

$$= x_1 [1 \quad -1 \quad -1]^T$$

$$\boxed{S_0 = \text{gen}\{[1 \quad -1 \quad -1]^T\}}$$

Busco $S_3 = \text{Nul} \begin{pmatrix} 1-3 & 1 & 0 \\ 1 & 2-3 & -1 \\ 0 & -1 & 1-3 \end{pmatrix} = \text{Nul} \begin{pmatrix} -2 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -2 \end{pmatrix}$

$$= \text{Nul} \begin{pmatrix} -2 & 1 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{cases} -2x_1 + x_2 = 0 & \rightarrow x_2 = 2x_1 \\ -x_2 - 2x_3 = 0 & \rightarrow -2x_1 - 2x_3 = 0 \end{cases}$$

$$2x_3 = -2x_1$$

$$\boxed{x_3 = -x_1}$$

$$[x_1 \quad x_2 \quad x_3] = [x_1 \quad 2x_1 \quad -x_1] = x_1 [1 \quad 2 \quad -1]$$

$$\boxed{S_3 = \text{gen}\{[1 \quad 2 \quad -1]^T\}}$$

$$B = \{ [1\ 0\ 1]^T\ [1\ -1\ -1]^T\ [1\ 2\ -1]^T \}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & -1 \end{pmatrix} \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix} \qquad D \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ -1 & 1 & -1 \end{pmatrix}^{-1}$$

Para calcular $A^{1961} x = [2\ 0\ 2]^T$

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

Aplico

$$A^{1961} x = \alpha_1 A^{1961} v_1 + \alpha_2 A^{1961} v_2 + \alpha_3 A^{1961} v_3$$

$$A^{1961} x = \alpha_1 (0)^{1961} v_1 + \alpha_2 (1)^{1961} v_2 + \alpha_3 (3)^{1961} v_3$$

$$A^{1961} x = \alpha_2 v_2 + \alpha_3 (3)^{1961} v_3$$

Sabemos que $A^{1961} x = [2\ 0\ 2]^T$

$$\alpha_2 v_2 + \alpha_3 (3)^{1961} v_3 = 0 v_1 + 2 v_2 + 0 v_3$$

$\alpha_1 \in \mathbb{R}$ $\qquad \alpha_2 = 2$ $\qquad \alpha_3 = 0$

La solución entonces es

$$X = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$$

$$X = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$X = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\boxed{X = \alpha_1 \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} \qquad \alpha_1 \in \mathbb{R}}$$

$\left[\begin{matrix}0 & -1 & 1\end{matrix}\right]$

3. Hallar una matriz simétrica $A \in \mathbb{R}^{3\times3}$ tal que $\begin{bmatrix}1 & 0 & 1\end{bmatrix}^T$ y $\begin{bmatrix}1 & 1 & 1\end{bmatrix}^T$ son autovectores de A, traza$(A) = 7$ y $\det(A) = 12$.

4. Sea $A \in \mathbb{R}^{2\times3}$ la matriz definida por

Sabemos que $[1\ 0\ 1]^T$ y $[1\ 1\ 1]^T$ son autovectores de A

$Tra(A) = 7$ y $\det(A) = 12$

$\langle [1\ 0\ 1][1\ 1\ 1] \rangle = 1 + 0 + 1 = 2$ $\therefore$ v_1 y v_2 tienen asociado el mismo autovalor

Construimos $v_3 = v_1 \times v_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{vmatrix} = [0-1 \quad 1-1 \quad 1]^T = [-1\ 0\ 1]^T$

Sabemos que existe un $\lambda \in \sigma(A)$ tal que $m(\lambda) = 2$

$\mu \in \sigma(A)$ tal que $m(\mu) = 1$

$Tra(A) = 7$ $\wedge$ $\det(A) = 12$

$\lambda + \lambda + \mu = 7$

$2\lambda + \mu = 7$

$\mu = 7 - 2\lambda$

$\lambda \cdot \lambda \cdot \mu = 12$

$\lambda^2 \cdot \mu = 12$

$\lambda^2(7 - 2\lambda) = 12$

$7\lambda^2 - 2\lambda^3 = 12$

$2\lambda^3 - 7\lambda^2 + 12 = 0$

aplicando $p(z) = 0$

$\lambda = 2$ cumple con la Ecuacion

$\mu = 7 - 2(2) = 7 - 4 = 3$

entonces $\lambda = 2$ con $m(2) = 2$ y $\mu = 3$ con $m(3) = 1$

Tenemos entonces

$$P = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Entonces

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{(-1)}$$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1/2 & -1 & 1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1/2 \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & 0 \\ -3/2 & 0 & 3/2 \end{pmatrix}$$

$$\boxed{A = \begin{pmatrix} 5/2 & 0 & -1/2 \\ 0 & 2 & 0 \\ -1/2 & 0 & 5/2 \end{pmatrix}}$$

traza$(A) = 7$ y $\det(A) = 12$.

4. Sea $A \in \mathbb{R}^{2\times 3}$ la matriz definida por

$$A = 4\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 0 & 1\end{bmatrix} + 2\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1 & 0 & -1\end{bmatrix}.$$

Hallar $A^\dagger$, la seudoinversa de Moore-Penrose de A, y determinar la solución por mínimos cuadrados de norma mínima de la ecuación $Ax = \begin{bmatrix}1 & 2\end{bmatrix}^T$.

5. Sea $T: \mathbb{R}^3 \to \mathbb{R}^2$ la transformación lineal definida por $T(x) = Ax$, donde

$$\langle [1\ 1]\,[1\ -1] \rangle = 0$$
$$\langle [1\ 0\ 1][1\ 0\ -1] \rangle = 0$$

$$A = 4\sqrt{2}\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}\sqrt{2}\begin{bmatrix}1/\sqrt{2} & 0 & 1/\sqrt{2}\end{bmatrix} + 2\sqrt{2}\begin{bmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{bmatrix}\sqrt{2}\begin{bmatrix}1/\sqrt{2} & 0 & -1/\sqrt{2}\end{bmatrix}$$

$$A = 4\sqrt{2}^2\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}\begin{bmatrix}1/\sqrt{2} & 0 & 1/\sqrt{2}\end{bmatrix} + 2\sqrt{2}^2\begin{bmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{bmatrix}\begin{bmatrix}1/\sqrt{2} & 0 & -1/\sqrt{2}\end{bmatrix}$$

$$A = 8\begin{pmatrix}1/\sqrt{2}\\1/\sqrt{2}\end{pmatrix}\begin{bmatrix}1/\sqrt{2} & 0 & 1/\sqrt{2}\end{bmatrix} + 4\begin{bmatrix}1/\sqrt{2}\\-1/\sqrt{2}\end{bmatrix}\begin{bmatrix}1/\sqrt{2} & 0 & -1/\sqrt{2}\end{bmatrix}$$

Podemos encontrar $U = \begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}$ $\quad \Sigma = \begin{pmatrix}8 & 0\\ 0 & 4\end{pmatrix}$

$$V^T = \begin{pmatrix}1/\sqrt{2} & 0 & 1/\sqrt{2}\\ 1/\sqrt{2} & 0 & -1/\sqrt{2}\end{pmatrix}$$

$$A = \begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}\begin{pmatrix}8 & 0\\ 0 & 4\end{pmatrix}\begin{pmatrix}1/\sqrt{2} & 0 & 1/\sqrt{2}\\ 1/\sqrt{2} & 0 & -1/\sqrt{2}\end{pmatrix}$$

$$A^+ = \begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 0 & 0\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}\begin{pmatrix}8 & 0\\ 0 & 4\end{pmatrix}^{(-1)}\begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}$$

$$A^+ = \begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 0 & 0\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}\begin{pmatrix}1/8 & 0\\ 0 & 1/4\end{pmatrix}\begin{pmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{pmatrix}$$

$$A^{+} = \frac{1}{16}\begin{pmatrix} 3 & -1 \\ 0 & 0 \\ -1 & 3 \end{pmatrix}$$

para Resolve $Ax = [1\ 2]^{T}$

Usamos $x = A^{+}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$$x = \frac{1}{16}\begin{pmatrix} 3 & -1 \\ 0 & 0 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$x = \frac{1}{16}\begin{pmatrix} 3 - 2 \\ 0 \\ -1 + 6 \end{pmatrix}$$

$$\boxed{x = \frac{1}{16}\begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}}$$